Table 1: Five-DGP comparison. Each cell reports validity (%, ↑) / joint MMD (↓) / NN coverage (%, ↑), averaged over three seeds.

| Method | Discrete | Linear | Cubic | Interaction | 2D Bridge |
|---|---|---|---|---|---|
| CD | 18.33 / 0.0896 / 44.45 | 83.22 / **0.0025** / **78.52** | 78.12 / 0.0062 / 74.35 | **84.98** / 0.0030 / 67.97 | 80.65 / 0.0072 / 73.08 |
| Compositional CD | **94.37** / 0.0747 / **76.73** | 38.54 / **0.0139** / 31.25 | 31.74 / 0.0368 / 27.67 | 6.84 / 0.0717 / 6.05 | 20.93 / 0.1054 / 20.04 |
| Additive mean† | 0.00 / 0.1937 / 0.00 | **100.00** / **0.0088** / 0.55 | 0.00 / 0.1711 / 0.00 | **100.00** / 0.0169 / **0.65** | 0.00 / 0.3416 / 0.00 |
| Additive + residual† | 2.80 / 0.2020 / **10.60** | **7.81** / **0.1407** / 6.17 | 0.00 / 0.2037 / 0.00 | 3.11 / 0.1431 / 1.76 | 0.18 / 0.3266 / 0.10 |
| Fixed composition + residual† | **3.45** / 0.2344 / **12.47** | 0.03 / 0.2327 / 0.00 | 0.02 / **0.2014** / 0.00 | 0.05 / 0.3248 / 0.00 | 0.00 / 0.3798 / 0.00 |
| Rep4Ex-inspired + residual† | 1.70 / **0.2076** / **5.52** | **3.68** / 0.2490 / 3.06 | 0.23 / 0.2732 / 0.02 | 1.04 / 0.2488 / 0.75 | 0.03 / 0.3648 / 0.03 |
| CG: sparse factorized ($s = 1$) | **83.57** / **0.0489** / **89.53** | 5.40 / 0.0776 / 1.84 | 8.92 / 0.1280 / 4.07 | 2.03 / 0.1874 / 0.76 | 4.07 / 0.2045 / 3.52 |
| CG: dense factorized ($s = 1$) | **88.45** / **0.0661** / **85.35** | 7.67 / 0.0692 / 2.86 | 3.45 / 0.1266 / 0.65 | 1.32 / 0.1780 / 0.73 | 1.69 / 0.2084 / 1.79 |
| CG: dense joint ($s = 1$) | 1.40 / 0.4381 / 3.93 | **15.35** / 0.0359 / **6.74** | 13.22 / 0.0745 / 5.49 | 13.13 / **0.0339** / 4.26 | 6.18 / 0.1360 / 4.82 |
| SEDGE: no weighting | 7.33 / **0.2352** / **22.30** | **8.98** / 0.4006 / 20.82 | 8.97 / 0.3650 / 17.61 | 8.53 / 0.4809 / 14.01 | 2.46 / 0.6294 / 4.10 |
| SEDGE-LGD ($s = 0.5$) | 45.28 / **0.0703** / **74.82** | 44.11 / 0.1343 / 64.81 | 31.92 / 0.2048 / 49.61 | **49.25** / 0.1531 / 53.34 | 23.67 / 0.3420 / 35.40 |
| SEDGE-LGD ($s = 1$) | 77.55 / 0.0653 / **89.88** | 78.24 / **0.0104** / 79.17 | 60.17 / 0.0711 / 64.49 | **78.96** / 0.0263 / 62.94 | 66.37 / 0.1153 / 66.60 |
| SEDGE-LGD ($s = 2$) | 69.52 / 0.0815 / **72.63** | **77.07** / 0.0302 / 62.42 | 71.89 / **0.0302** / 50.91 | 70.36 / 0.0457 / 42.33 | 76.25 / 0.0516 / 56.77 |
| SEDGE-LWS | 88.40 / 0.0624 / **88.92** | 91.98 / 0.0018 / 87.37 | 91.26 / 0.0024 / 84.29 | **92.97** / 0.0039 / 79.12 | 87.39 / **0.0016** / 78.86 |

Bold identifies each method's best DGP separately for each metric (requested row-wise convention). This does not rank methods; MMD magnitudes across distinct DGPs need not be comparable. Validity requires both specifications and root support. Joint MMD compares $(X, Z)$ distributions; NN coverage measures the fraction of reference points covered at the original fixed-radius criterion. †Paper-inspired adaptations, not faithful reproductions. The additive mean is a deterministic control, not a posterior sampler. All three previously completed LGD scales are shown; CG rows fix $s = 1$. The expanded LGD sweep is not included. The discrete reference has 5% conditional exclusion and is an explicitly relaxed Assumption-3 control; The four continuous settings pass the working 1% criterion. Results are an exploratory selection; all-method, per-seed data are retained.